

Arseniy Shelonin

Candidate to Software Engineer
19 years old



Contact

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Achievements

Physics Olympiad Winner (11th grade)

🏆 All-Russian physics Olimpiad, Regional Stage
🏆 Kurchatov Olympiad
🏆 Engineering Olympiad
🏆 All-Siberian Olympiad

Education

Moscow Institute of Physics and Technology (MIPT)

Department of Radio Engineering and Cybernetics,
Bachelor's program, 1st year
GPA: 7.14/10

2024–present

Skills

IT	C, x86-64 assembly, LaTeX
Tools	git, make, GDB, radare2, IDA, SFML, TXXLib, Graphviz, Gnuplot
Engineering	Solidworks, Autodesk Fusion 360, Blender, EasyEda
Hobbies	Circuit design, Arduino
Languages	Russian (native), English (B1)

Technical Projects

Punk Language Compiler 🌐 github.com/PUNK_LANGUAGE C, Graphviz, GDB, Make

- A programming language project that uses punk-inspired lexicon.
- Implemented three main components: frontend, middle-end, and backend.
The frontend builds an abstract syntax tree from code; the middle-end performs basic optimizations; the backend generates machine code for my custom assembler.

Virtual Processor

 github.com/SPU

C, Graphviz, TXLib, Make

- A project simulating CPU and assembler operation.
- The assembler translates assembly code into binary format.
- The processor executes the machine code generated by the assembler.
- Implemented register, stack, and RAM operations.

Differentiator

 github.com/DIFFERENTIATOR

C, Graphviz, Gnuplot, GDB, Make

- A program that calculates nth-order derivatives of arbitrary expressions and expands them into Taylor series up to $o(n)$.
- The program simplifies expressions before each differentiation and simplifies the results.
- Finally, it generates LaTeX code and a PDF file with step-by-step explanations and plots the Taylor approximation polynomial.

Mandelbrot Set Calculator

 github.com/FRACTAL

C, make, SFML

- Implemented 4 versions of the calculation algorithm. The simplest version calculates each pixel without utilizing all CPU resources. The optimized version uses SIMD instructions and loop unrolling.

Printf Implementation

 github.com/PRINTF

x86-64 assembly, radare2

- Implementation of the standard library printf function following System V AMD64 calling conventions.
- Features floating-point numbers.

Crack

 github.com/CRACK

C, x86-64 assembly, radare2

- The project consists of two parts: a vulnerable program and a crack - a program that cracks the partner's program.
- The crack features a graphical interface and works with the partner program's binary code.